

Unit economics of \$10K

Week 8, Lecture 1 · LangoBee to \$10K MRR

Summer 2026 · Operating a Solo Language App from Zero Revenue

What we'll cover

- How many subscribers \$10K MRR actually requires at \$7.99/mo and at \$49/yr
- Churn as a compounding tax: the math of the churn ceiling
- The Quick Ratio: a one-number growth health check
- LTV at education-category benchmarks
- What LTV implies for the maximum CAC you can defend

The subscriber count

\$10K MRR at LangoBee's prices

Three configurations, three answers:

- **Monthly plan only (\$7.99/mo):** $\$10,000 / \$7.99 = 1,252$ subscribers
- **Annual plan only (\$49/yr = \$4.08/mo):** $\$10,000 / \$4.08 = 2,451$ subscribers
- **60/40 blend (60% annual, 40% monthly):** blended ARPA = \$5.63/mo; $\$10,000 / \$5.63 = 1,776$ subscribers

The annual plan grows the catalog of buyers you need by 96% compared to monthly-only.

Pricing the annual plan as the prominent option is the single lever that most changes this arithmetic.

(Skok, 2020; LangoBee June 2026 snapshot)

Why the blend matters

RevenueCat's 2025 report: for education apps, annual plan year-1 retention is 44.1%, versus much lower for monthly plans.

An annual subscriber at \$49 contributes \$49 upfront and stays 44% through year 2. A monthly subscriber at \$7.99 contributes only if they stay each month.

The annual plan has two effects on the model:

1. Raises subscriber count needed (more people at lower ARPA)
2. Reduces effective churn (annual plans churn at the renewal date, not monthly)

Neither effect is obviously better. The model tells you which one dominates.

(RevenueCat Research Team, 2025)

Churn as a compounding tax

The churn ceiling formula

Every subscription business has a ceiling:

```
Steady-state subscribers = New subscribers/month / Monthly churn rate
```

At 5% monthly churn and 63 new subscribers per month:

```
Ceiling = 63 / 0.05 = 1,260 subscribers  
MRR ceiling at $7.99 = 1,260 × $7.99 = ~$10,067
```

So 63 new subscribers per month at 5% churn barely clears \$10K MRR at the monthly price. Any fewer new subscribers, or any higher churn, and the ceiling drops below the target.

(Rachitsky, 2022; Skok, 2020)

What 5% monthly churn feels like

5% monthly churn means roughly:

- 1 in 20 subscribers cancel each month
- After 12 months, about 46% of your original cohort has churned
- After 24 months, about 71% has churned

Compare Rachitsky (2022) benchmarks:

- Good B2C: 3-5% monthly churn
- Great B2C: under 2% monthly churn

At 2% monthly churn, 63 new subs/month produces a ceiling of 3,150 subscribers and \$25K+ MRR. The ceiling triples when churn halves.

(Rachitsky, 2022)

Why churn is a compounding tax

The analogy is compound interest in reverse. Every month, a percentage of your subscriber base disappears before the month's new revenue is counted.

```
Month 1: 0 subs + 63 new - 0 churn = 63
Month 2: 63 + 63 new - 3 churn = 123
Month 3: 123 + 63 new - 6 churn = 180
...
Month 24: approaching ceiling, ~1,260
```

Adding more traffic does not change the ceiling. Only reducing churn or increasing the new-sub rate changes it.

This is why week 7's retention work is not a prelude to growth. It is the prerequisite.

(Walling, 2022)

Quick Ratio and LTV

The Quick Ratio: one health number

Quick Ratio = (New + Resurrected subscribers) / Churned subscribers

- Above 4: growing fast and efficiently
- 2-4: healthy growth
- 1-2: barely growing; churn is nearly matching new acquisition
- Below 1: shrinking

At \$0 MRR today, LangoBee's Quick Ratio is undefined. The first subscribers will make it calculable. The first target: keep Quick Ratio above 2 through months 1-6.

LTV at education benchmarks

LTV (lifetime value) = ARPA / Monthly churn rate

At \$7.99/mo and 5% monthly churn: $LTV = \$7.99 / 0.05 = \textbf{\$159.80}$

RevenueCat 2025 reality check for education apps:

- Median Year 1 LTV per payer: **\$10.97**
- P90 Year 1 LTV: **\$52.41**

The formula-derived LTV assumes subscribers stay indefinitely. The RevenueCat median reflects observed behavior: most app subscribers churn within year 1. Use \$10.97-\$52.41 as the planning range, not \$159.80.

(RevenueCat Research Team, 2025; Skok, 2020)

Maximum defensible CAC

Skok (2020): LTV:CAC should exceed 3:1. Months to recover CAC should be under 12.

At the RevenueCat median LTV of \$10.97:

$$\text{Max CAC} = \$10.97 / 3 = \$3.66$$

At the P90 LTV of \$52.41:

$$\text{Max CAC} = \$52.41 / 3 = \$17.47$$

At the formula-derived LTV of \$159.80 (optimistic):

$$\text{Max CAC} = \$159.80 / 3 = \$53.27$$

Which LTV to use depends on churn. You do not know your churn yet. Use \$3.66 as the floor (conservative), \$17.47 as the base case.

Month 1-3 churn is typically activation failure, not product-market fit decay. Cohort-based analysis surfaces this pattern; blended monthly rates hide it. (Paraphrased)

Lenny Rachitsky (2022)

What Is Good Monthly Churn

What this means for LangoBee right now

LangoBee has \$0 MRR and zero trial starts (June 2026 snapshot). That means:

- Every number in this model is assumed, not measured
- The most dangerous assumption is trial-to-paid: no data exists
- The churn ceiling is irrelevant until there are subscribers to churn

The immediate move is not to optimize the model. It is to get the first 10 trial starts so you have one real data point in the funnel.

(LangoBee June 2026 snapshot)

Take-aways

1. \$10K MRR requires 1,252 monthly subscribers or 2,451 annual subscribers; a 60/40 blend lands at 1,776.
2. The churn ceiling formula (new subs / churn rate) sets an absolute growth cap that more traffic cannot break.
3. At 5% monthly churn, 63 new subscribers per month barely reaches \$10K MRR. Halving churn to 2% triples the ceiling.
4. Education-category median LTV is \$10.97 per payer (RevenueCat 2025), implying a maximum defensible CAC of \$3.66-\$17.47 depending on churn assumptions.
5. LangoBee has no trial data. The model is arithmetic until the first 10 trial starts replace assumptions with measurements.

What's next

- **Reading:** "Week 8: The \$10K math" at </c/langobee-10k-mrr-26su/readings/wk08>
- **Lecture 2 (this week):** Working backward to traffic: three channel scenarios and the funnel arithmetic
- **Section this week:** Build the MRR model spreadsheet; stress-test all three scenarios
- **HW3 due this week.** Capstone "The \$10K MRR Operating Plan" goes out this week (due week 12).